Crushing Time Lab

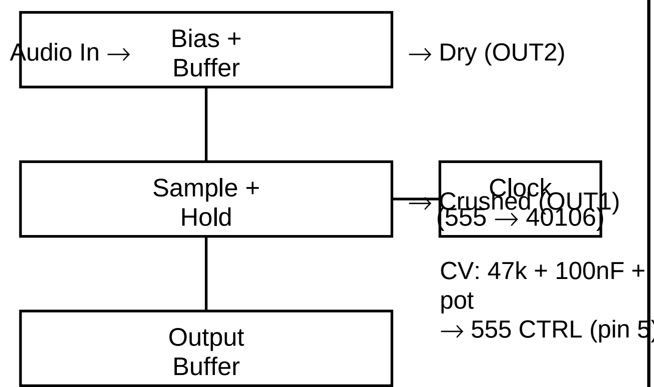
AE Modular BRAEDBOARD • Voltage-controlled sample-rate reducer ("bit-crusher")

Core idea: a clock samples audio too slowly. Slow = steps/freeze. Fast = rasp/aliasing.
Single-supply = build Vref (~2.5V). Outputs: OUT2 dry, OUT1 crushed. CV bends the crush rate (attenuate + filter, inject to 555 CTRL pin 5).

Build in layers

- 1) Rails + Vref: 100k to +5, 100k to GND, 100nF to GND (Vref ~2.5V).
- 2) Bias + buffer (MCP602): AC-couple in, bias to Vref, follower -> AUDIO_BIASED.
- 3) Clock (NE555): astable with rate pot; add decoupling on +5 near the chip.
- 4) Clean clock (40106): CLK_RAW -> Schmitt -> CLK.
- 5) Sample/Hold (4051): X0=AUDIO_BIASED, Z=SAMP_NODE, hold cap to Vref; INH=CLK.
- 6) Output buffer (MCP602): follower -> OUT1. Dry buffer -> OUT2.

Concept map



Checkpoints (measure + listen)

- Vref: 2.5V ± 0.2V (stable).
- CLK: 0-5V, rate sweeps with pot.
- OUT2: dry passes audio; idle near Vref.
- OUT1: slow=steps, fast=grit.
- Ticking? Improve decoupling, shorten CLK lead, move hold cap closer.

CV-controlled crush (wiring)

- CV socket → 47k → CV_IN; 100nF from CV_IN to GND.
- CV Amount pot: CV_IN to pot, other end GND, wiper = CV_ATTEN.
- CV_ATTEN → 100k → 555 pin 5 (CTRL).
- At pin 5: 10nF to GND + clamp diodes (1N4148) to +5 and GND.

Patch recipes + hygiene

1) Classic: source → CV (audio in). OUT1 → VCA/mix. OUT2 → mix for dry blend.
2) Breathing: LFO → CV (rate mod). Keep CV Amount low; raise until musical.
3) Sync: patch AE CLK to clock input (replace internal clock) to lock to tempo.
Hygiene: decouple each IC; keep CLK away from SAMP_NODE; hold cap close to CD4051 Z pin